MEDAITHON

CODE 4 CURE



PROBLEM STATEMENTS 2026

MED-
PS-001

1.Cardiology

Problem Statement: Development of a Motion-Compensated Wearable System for Detecting Transient Cardiac Events

Brief cardiac rhythm disturbances may be missed by intermittent measurements, while movement-related noise can reduce the reliability of wearable ECG recordings.

Proposed Innovation

- A wearable ECG sensor that continuously records the user's heart activity.
- A motion sensor that detects body movements and physical activity during ECG recording.
- A signal-quality analysis module that identifies noise and unreliable ECG measurements.
- An event detection engine that identifies potentially significant transient cardiac events.
- A signal correlation mechanism that combines ECG patterns, motion data, and signal quality to reduce false alerts.
- An explainable alert system that highlights important events and provides recordings for clinical review.

Potential patent focus: A sensor-fusion and adaptive signal-reconstruction mechanism that responds to different types of motion artefacts.

MED-

PS-001

REFERENCE

	Patent	Relevance to your concept	Link
1	**US10524736B2 – System and Method to Detect Significant Arrhythmic Events Through a Photoplethysmogram (PPG) and Accelerometer**	Uses a separate motion sensor to judge signal validity and reject motion artefacts before flagging significant arrhythmic events (PPG-based rather than ECG).	Google Patents – US10524736B2
2	**US20140378848A1 – Method and Apparatus for Motion Artifact Reduction in ECG Harness**	ECG harness with motion-artefact reduction components for the electrodes, aimed at clinical-quality ECG during movement.	Google Patents – US20140378848A1
3	**US9955887 – Wearable Monitor**	Wearable ECG monitor that compares the accelerometer frequency spectrum with the ECG signal and marks matching segments as artefact, improving arrhythmia detection.	Google Patents – US9955887
4	**US11605458 – Wearable Monitor**	Same wearable ECG-plus-accelerometer line: activity-aware arrhythmia detection and free-fall (fainting) triggers that start data transmission.	Google Patents – US11605458

5	**US12213791 – Wearable Device**	Later member of the same wearable ECG and motion-sensing disclosure; review its claims for how different artefact types are handled.	Google Patents – US12213791

2. Neurology

Problem Statement: Development of a Contactless Multimodal System for Tracking Subtle Neuromotor Changes

Small changes in gait, hand movement, facial motion, or speech may go unnoticed between neurological assessments, particularly in patients requiring longitudinal monitoring.

Proposed Innovation

- A camera-based monitoring system that records patient movements during standardized assessment tasks.
- A microphone module that captures speech patterns to assess changes in voice and speech characteristics.
- A patient-specific baseline model that establishes normal movement and speech patterns.
- A movement and speech analysis engine that identifies deviations from the patient's baseline.
- A change detection mechanism that tracks variations in movement, coordination, and speech over time.
- A clinician review dashboard that presents assessment results and detected changes to support clinical evaluation.

Potential patent focus: A synchronized multimodal calibration method that compensates for camera position, lighting, and differences in task execution.

REFERENCE

	Patent	Relevance to your concept	Link
1	**US8698888B2 – Systems and Methods for Comprehensive Human Movement Analysis**	Very close prior art: single camera with configurable camera settings, audio-instructed movement-examination modules and automatic measurement of the patient's recorded performance.	Google Patents – US8698888B2
2	**US11986286B2 – Gait-Based Assessment of Neurodegeneration**	Gait-based assessment for neurodegenerative and movement disorders; classified under movement-disease monitoring and image/marker-based sensing.	Google Patents – US11986286B2
3	**US12336809B2 – Contactless Patient Motion Monitoring**	Contactless monitoring of patient movement as a vital sign, addressing gaps when caregivers are not at the bedside.	Google Patents – US12336809B2
4	**US20200289027A1 – System, Method and Computer Program Product for Assessment of a User's Gait**	Gait monitoring that selects the gait-analysis process according to the capture device, relevant to compensating for setup differences.	Google Patents – US20200289027A1
5	**US12279882 – Movement Disorder Diagnostics from Video Data Using Body Landmark Tracking**	Phone-camera video of standardized tasks (finger tapping, speech, facial expression, gait) analysed as biomarkers; closest to your standardized-task approach.	Google Patents – US12279882

	Research	What it supports	Link
1	Smartphone automated motor and speech analysis for early detection of Alzheimer's disease and Parkinson's disease: Validation of TapTalk across 20 different devices	Validates smartphone video and audio tasks (finger tapping, speech) against gold-standard sensors across 20 phones; supports your device and setup variation concern.	PMC – TapTalk validation
2	Contactless Gait Assessment in Home-like Environments (Sensors, 2021)	Shows a contactless LiDAR gait system correlating strongly (r>0.95) with pressure-mat and IMU references; supports contactless longitudinal gait tracking.	BORIS – Contactless gait assessment

MEDAITHON
CODE 4 CURE
2026

MED-PS-003

3.Endocrinology

Problem Statement: Development of a Personalized Glucose-Response Monitoring System That Accounts for Medication Timing and Daily Activity

Glucose responses vary with meals, activity, sleep, medication timing, and individual physiology. Generalized alerts may not adequately reflect a patient's recurring patterns.

Proposed Innovation

- A continuous glucose monitoring system that records blood glucose levels over time.
- A meal-tracking module that records food intake and meal timings.
- An activity sensor that monitors physical activity and movement patterns.
- A medication tracking system that records dosage and medication schedules.
- A pattern analysis engine that correlates glucose levels with meals, activity, and medication to identify recurring trends.
- A clinician-reviewed dashboard that presents glucose patterns and personalized insights to support diabetes management.

Potential patent focus: A technical method for resolving conflicting or missing contextual inputs when estimating individualized glucose-response patterns.

MEDATHON
CODE 4 CURE
2026

MED-PS-003

REFERENCE

	Patent	Relevance to your concept	Link
1	**US11553883 – System, Device and Method of Dynamic Glucose Profile Response to Physiological Parameters**	Very close prior art: multi-phase glucose-response patterns personalised to activity, meal and medication intake and updated dynamically as new data arrives.	Google Patents – US11553883
2	**US11717225B2 – Method and Apparatus for Determining Meal Start and Peak Events in Analyte Monitoring Systems**	Infers meal start and peak from dense CGM data combined with user meal markers; addresses variable meal timing.	Google Patents – US11717225B2
3	**US8756043B2 – Blood Glucose Meter and Computer-Implemented Method for Improving Glucose Management Through Modeling of Circadian Profiles**	Stores pre- and post-meal glucose and medication-dose data per meal period as circadian profiles.	Google Patents – US8756043B2
4	**US20090240127A1 – Methods of Determining Pre or Post Meal Time Slots or Intervals in Diabetes Management**	Identifies glucose patterns (hypo/hyperglycaemic, variability) from glucose and insulin data and shows warnings.	Google Patents – US20090240127A1
5	**US12283381B2 – Systems and Methods for Continuous Glucose**	Combines CGM data with medication, diet and activity engagement data in a machine-	Google Patents – US12283381B2

	Monitoring Outcome Predictions	learning model that	

		drives timing and dosing notifications.	

4.Gastroenterology

Problem Statement: Development of a Sensor-Based Digestive Transit and Medication Adherence Monitoring Platform

Patients with gastrointestinal motility disorders may experience variable symptoms, while clinical teams often lack objective, continuous information about symptom timing and digestive transit.

Proposed Innovation

· A wearable sensor system that continuously records physiological data such as heart rate, temperature, and activity levels.

· A structured symptom-recording module that captures symptom types, severity, and occurrence times.

· A medication-event logging system that records medication intake and dosage timings.

· A data correlation engine that identifies relationships between physiological changes, reported symptoms, and medication events.

· A pattern detection mechanism that identifies recurring trends associated with reported symptoms.

· A research dashboard that presents findings and timelines for clinician review and further investigation.

Potential patent focus: A sensor-calibration and event-correlation mechanism that aligns physiological measurements with patient-reported gastrointestinal events.

MEDAITHON
CODE 4 CURE
2026

MED-PS-004

REFERENCE

	Patent	Relevance to your concept	Link
1	**US11864907B2 – Devices, Systems, and Methods for Monitoring Gastrointestinal Motility**	Sensor-based GI motility monitoring that compares signal characteristics against thresholds indicating a GI condition.	Google Patents – US11864907B2
2	**US10231692B2 – Abdominal Statistics Physiological Monitoring System and Methods**	Acoustic sensor registers digestive motility events and rejects noise-induced signals; close to wearable event detection.	Google Patents – US10231692B2
3	**US6056703A – Method and Apparatus for Characterizing Gastrointestinal Sounds**	Torso microphone array with ambulatory recording and cross-channel event timing for GI sounds.	Google Patents – US6056703A
4	**US20100172839A1 – Method and System for Evaluating Gastrointestinal Motility**	Acoustic sensors to estimate GI events and transit time. Google Patents lists this application as abandoned, but it remains citable prior art.	Google Patents – US20100172839A1
5	**US9028407 – Methods and Apparatus for Monitoring Patient Conditions**	Abdominal accelerometer captures bowel-sound vibrations; response to a prokinetic medication is used to judge motility, linking medication events to motility signals.	Google Patents – US9028407

5.Nephrology

Problem Statement: Development of a Closed-Loop Fluid-Balance Reconciliation and Anomaly Detection Device

Fluid intake, urine output, drain losses, and intravenous fluids may be recorded through different workflows, creating discrepancies that affect clinical fluid-balance assessments.

Proposed Innovation

- A smart collection container equipped with sensors to monitor fluid collection and disposal.
- Calibrated weight or flow sensors that measure fluid quantities accurately.
- A fluid administration logging system that records the volume and timing of fluid intake and output.
- A data validation engine that compares sensor readings with recorded entries.
- An anomaly detection mechanism that identifies missing, duplicated, or physically inconsistent records.
- A monitoring dashboard that highlights discrepancies and provides traceable reports for healthcare staff review.

Potential patent focus: A cross-sensor reconciliation mechanism that detects recording errors and estimates measurement uncertainty in real time.

MEDAITHON
CODE 4 CURE
2026

MED-PS-005

REFERENCE

	Patent	Relevance to your concept	Link
1	**US12246146B2 – Automated Weight Based Fluid Output Monitoring System**	Weight-based fluid-collection monitoring with a sensor array that detects pressure and angle changes between hanger and stand.	Google Patents – US12246146B2
2	**US9995619B2 – Fluid Container Measurement System Employing Load Cell Linkage Member**	Load-cell monitoring of urine or IV bags to replace manual charting of output.	Google Patents – US9995619B2
3	**US11911160B2 – Automated Urine Output Measurement Systems and Methods Thereof**	Load-cell urine monitor with display and electronic-record integration for ICU fluid-balance assessment.	Google Patents – US11911160B2
4	**WO2000037129A1 – Method and Device for Monitoring Fluid Level**	Monitors both IV fluid delivery and urine excretion with level and rate sensors to automate fluid-balance control.	Google Patents – WO2000037129A1
5	**US8075513B2 – Patient Connection System for a Balance Hydration Unit**	Weighs the urine bag and the hydration bag and controls infusion from measured urine output, a closed-loop example.	Google Patents – US8075513B2

MEDAITHON
CODE 4 CURE
2026

MED-PS-006

6.Pulmonary (Respiratory) Medicine

Problem Statement: Development of a Respiration Monitoring System That Separates Breathing Deterioration from Movement Artefacts

Wearable respiratory monitoring can become unreliable during walking, coughing, speaking, or changes in body posture, making it difficult to interpret changes in breathing patterns.

Proposed Innovation

- A respiratory motion sensor that monitors chest movements and breathing rate.
- An acoustic sensor that captures breathing sounds to identify changes in respiratory patterns.
- A body movement sensor that tracks posture and physical activity.
- A signal-quality analysis module that identifies noise and unreliable measurements.
- A multi-sensor analysis engine that combines respiratory, acoustic, and movement data to detect changes in breathing patterns.
- A monitoring dashboard that displays respiratory trends and flags unusual patterns for clinical review.

Potential patent focus: An adaptive signal-quality algorithm that selects or weights sensor channels according to activity and signal reliability.

MEDAITHON
CODE 4 CURE
2026

MED-PS-006

REFERENCE

	Patent	Relevance to your concept	Link
1	**US10052048B2 – Respiratory Monitors and Monitoring Methods**	Very close prior art: synthesises one respiratory signal from respiration-related motion and respiration-related sound, for robustness to motion and positioning.	Google Patents – US10052048B2
2	**US11857309B2 – Respiration Monitoring Device and Methods for Use**	Accelerometers plus an acoustic sensor for breath sound; classified under motion-artefact reduction using a separate motion sensor.	Google Patents – US11857309B2
3	**US9675282B2 – Method and Apparatus for Determining a Respiration Signal**	Multi-axial accelerometer respiration signal with adaptive noise cancellation to remove non-respiratory motion.	Google Patents – US9675282B2
4	**US8979765B2 – Body-Worn Monitor for Measuring Respiratory Rate**	Chooses the processing algorithm by degree of motion (peak counting, adaptive filtering, FFT).	Google Patents – US8979765B2
5	**US10617325 – Respiratory Rate Measurement Using a Combination of Respiration Signals**	Weights accelerometer- and ECG-derived respiratory rates by quality metrics so an unreliable channel contributes little; close to channel weighting.	Google Patents – US10617325

MEDATHON
CODE 4 CURE
2026

MED-PS-006

7. Oncology (Medical Oncology)

Problem Statement: Development of a Treatment-Cycle Symptom and Toxicity Monitoring System With Patient-Specific Escalation Rules

Symptoms arising between oncology visits may be underreported or recorded inconsistently, making it harder for clinical teams to recognize worsening toxicity promptly.

Proposed Innovation

- A structured symptom-reporting module that records symptom type, severity, and frequency.
- A treatment-cycle tracking system that monitors treatment stages and patient progress over time.
- A medication-recording module that maintains dosage, timing, and medication history.
- A threshold-based monitoring engine that compares reported symptoms with clinician-defined escalation criteria.
- An alert generation system that flags concerning symptoms or changes requiring medical attention.
- A patient-friendly dashboard that displays symptom trends, treatment timelines, and alerts for clinician review.

Potential patent focus: A time-aware event-processing mechanism that associates symptoms with treatment cycles, identifies conflicting reports, and prioritizes clinically configured alerts.

- **REFERENCE**

	Patent	Relevance to your concept	Link
1	**US20100191073A1 – System for Monitoring Chemotherapy-Associated Adverse Drug Reactions**	Very close prior art: patient-held terminal for periodic entry of predetermined adverse reactions during chemotherapy cycles, with server-generated alerts.	Google Patents – US20100191073A1
2	**US11450432 – Predictive and Interactive Diagnostic System**	Mobile symptom and severity reporting for cancer patients, contextualised with treatment type and medications.	Google Patents – US11450432
3	**US20200143917A1 – Systems and Methods for Monitoring and Managing Cancer Patients Risk for Acute Care Utilization and/or for Improving Treatment Tolerability**	Remote monitoring, tracking of patient-reported symptoms and support interventions for outpatient cancer treatment.	Google Patents – US20200143917A1
4	**US12232852 – Vascular Access Devices, Systems, and Methods for Monitoring Patient Health**	Monitoring of chemotherapy patients through a vascular access device, motivated by delays and gaps in questionnaire-based symptom reporting.	Google Patents – US12232852

	Research	What it supports	Link
1	Evaluation of an online platform for cancer patient self-reporting of chemotherapy toxicities	Online reporting platform with real-time alerts to clinicians for serious toxicities; alerts led to delayed treatment, dose changes and rescheduling.	PMC – Online toxicity self-reporting
2	Usability Evaluation of a Mobile Phone–Based System for Remote Monitoring and Management of Chemotherapy-Related Side Effects (ASyMS)	Mobile PROM system with a clinician alerting component for chemotherapy toxicity at home.	PMC – ASyMS usability
3	Patients' and Clinicians' Experiences Using a Real-Time Remote Monitoring System for Chemotherapy Symptom Management (ASyMS): Qualitative Study	Describes tiered alerts (amber within 8 hours, red within 30 minutes), supporting clinician-defined escalation thresholds.	PMC – ASyMS experiences
4	Automated home monitoring and management of patient-reported symptoms during chemotherapy: results of the symptom care at home RCT	Randomised trial of daily automated symptom reporting with threshold-based alerts to a nurse practitioner.	PMC – Symptom Care at Home RCT
5	Active surveillance of chemotherapy-related symptom burden via electronic patient-reported outcomes and sensor-enabled vital signs	Combines ePRO-CTCAE reporting and patient-reported vitals with back-end threshold alerts reviewed by advanced practitioners.	PMC – MD Anderson RPM pilot

	capture: protocol for a decentralised feasibility pilot study		

MED-
PS-08

8.Rheumatology

Problem Statement: Development of a Wearable Joint-Mobility and Inflammation-Symptom Correlation System

Joint symptoms fluctuate throughout the day, while clinic-based assessments may not capture variations in movement, stiffness, and functional limitation during everyday activities.

Proposed Innovation

- A wearable motion sensor system that tracks joint movement and range of motion.
- A structured symptom assessment module that records pain, stiffness, and discomfort levels.
- A movement analysis engine that measures changes in joint mobility and movement patterns over time.
- A data correlation mechanism that links sensor readings with reported symptoms.
- A trend detection system that identifies recurring stiffness patterns and changes in joint function.
- A monitoring dashboard that presents movement trends and symptom reports for clinician review.

Potential patent focus: A joint-specific sensor calibration and movement-segmentation method that differentiates routine activity from standardized functional assessment tasks.

MED-PS-008

REFERENCE

	Patent	Relevance to your concept	Link
1	**WO2017186832A1 – Monitoring Device for Patients with Joint Afflictions**	Joint-worn monitoring device with temperature, motion, humidity and pressure sensors for rheumatoid arthritis and osteoarthritis.	Google Patents – WO2017186832A1
2	**US10646157B2 – System and Method for Measuring Body Joint Range of Motion**	Converts IMU orientation data into clinically defined joint rotation, with user-defined limits and targets.	Google Patents – US10646157B2
3	**US12029940B2 – Single Sensor Wearable Device for Monitoring Joint Extension and Flexion**	Single wearable sensor for joint extension and flexion (knee-focused) with a visual display.	Google Patents – US12029940B2

	Research	What it supports	Link
1	Review of Wearable Sensor-Based Health Monitoring Glove Devices for Rheumatoid Arthritis (Sensors, 2021)	Reviews sensing, calibration and stiffness measurement in RA gloves; supports the need for objective joint-stiffness measurement.	PMC – RA glove review
2	Accuracy of a Custom Physical Activity and Knee Angle Measurement Sensor System for Patients with	Inertial system that classifies activity and measures knee angle together; supports separating activity from joint	PMC – Activity and knee-angle sensor

	Neuromuscular Disorders	range of motion.	

	and Gait Abnormalities (Sensors, 2015)		
3	Textile sensor glove for health monitoring – Application in home assessment of Rheumatoid Arthritis (Coyle et al., 2013)	Fabric stretch-sensor glove for home assessment of joint stiffness through range and velocity of movement.	DORAS – Textile sensor glove

MEDAITHON
CODE 4 CURE
2026

MED-PS-09

09.Infectious Diseases

Problem Statement: Development of a Smart Specimen-Chain System for Detecting Temperature and Handling Deviations in Infectious Disease Diagnostics

Diagnostic specimens can be compromised by transport delays, incorrect temperatures, unsuitable handling, or incomplete chain-of-custody records.

Proposed Innovation

- A sensor-enabled specimen container that monitors environmental conditions during sample storage and transportation.
- A seal-status monitoring system that detects unauthorized opening or accidental seal damage.
- A transit-tracking module that records sample transfer events and transportation history.
- A time-logging system that tracks the duration between specimen collection and laboratory delivery.
- A quality-risk assessment engine that identifies temperature deviations, seal breaches, and transportation delays.
- A quality-reporting dashboard that generates a specimen risk report before laboratory processing.

Potential patent focus: A container-and-reader system that combines multiple exposure events into a specimen-specific integrity assessment.

MED-PS-009

REFERENCE

	Patent	Relevance to your concept	Link
1	**US11977088 – History Logging for Samples of Biological Material**	Very close prior art: transport container with environment sensors, handling and transit-time history, and comparison against test-specific delay thresholds to grade conformity or non-conformity.	Google Patents – US11977088
2	**EP3478413A1 – Monitoring Apparatus for Temperature-Controlled Sample Collection and Transport**	Temperature monitoring apparatus for sample transport (US family member: US11589576B2).	Google Patents – EP3478413A1
3	**EP3673450A1 – Intelligente Verfahren zur Überprüfung, Überwachung und Verwaltung von Proben (smart methods for checking, monitoring and managing samples)**	Checks sample-container parameters from filling to transport, records a temperature history log and tracks samples in real time.	Google Patents – EP3673450A1
4	**US6978891B2 – Diagnostic Specimen Transport Packaging and Methods of Use**	Packaging for diagnostic specimens that may contain infectious agents (e.g., HIV) under extreme-temperature transport and containment-failure risks.	Google Patents – US6978891B2
5	**US6769544B2 – Containment Envelope**	Baseline art on containment and sealing of infectious	Google Patents – US6769544B2

	for Diagnostic Specimens	diagnostic specimens during cold transport.	

MED-PS-009

10. Pediatrics

Problem Statement: Development of an Adaptive Pediatric Measurement System for Detecting Unreliable Vital-Sign Readings During Movement

Obtaining reliable measurements from infants and young children can be difficult because of movement, crying, poor sensor contact, and changing body size.

Proposed Innovation

- A pediatric monitoring device that records vital signs using age-appropriate measurement protocols.
- A contact-quality sensor that checks proper device placement and sensor contact.
- A motion detection module that identifies movement-related disturbances during measurements.
- A signal-quality analysis engine that determines whether readings are reliable or require repetition.
- An age-specific validation system that evaluates measurements against appropriate pediatric reference ranges.
- A monitoring dashboard that displays validated readings and flags unreliable measurements for healthcare professional review.

Potential patent focus: A size-adaptive sensor positioning and measurement-validation mechanism for different pediatric age groups.

REFERENCE

	Patent	Relevance to your concept	Link
1	**US10085639B2 – Tracking Contact Quality to Vital Signs Measurement Sensors**	Derives a contact-quality indication from sensor gain behaviour to flag poor-contact or movement-affected pulse-oximeter readings.	Google Patents – US10085639B2
2	**US7190987B2 – Neonatal Bootie Wrap**	Holder that positions a pulse-oximeter sensor at selectable locations on a neonate's extremities and reduces motion artefact; relevant to size-adaptive placement.	Google Patents – US7190987B2
3	**US7650177B2 – Medical Sensor for Reducing Motion Artifacts and Technique for Using the Same**	Sensor-level approach to motion artefacts in pulse oximetry, including movement by caregivers, agitation and shivering.	Google Patents – US7650177B2
4	**US8971979B2 – Medical Sensor for Reducing Motion Artifacts and Technique for Using the Same**	Continuation: sensor with a motion-damping structure (e.g., a dashpot) to damp forces on the sensor body.	Google Patents – US8971979B2
5	**US8260391B2 – Medical Sensor for Reducing Motion Artifacts and Technique for Using the Same**	Continuation: motion-damping sensor designs, including a bandage-style variant.	Google Patents – US8260391B2

	Research	What it supports	Link
1	A computer algorithm for differentiating valid from distorted pulse oximeter waveforms in neonates (Pediatric Pulmonology, 1995)	Algorithm that recognises valid versus motion-distorted pulse waveforms in neonates (92% sensitivity and positive predictive value in the abstract).	Neonatology on the Web – Adams et al.
2	Rejection of false saturation data in optical pulse oximeter (SPIE 7715)	Real-time validation that rejects SpO2 values caused by motion or poor skin contact in NICU recordings and substitutes short rejected gaps.	SPIE – Rejection of false SpO2 data

MED-
PS-011

11.Emergency Medicine

Problem Statement: Development of an Adaptive Emergency Triage Queue System That Reconciles Clinical Urgency With Resource Availability

Emergency departments must prioritize patients while balancing changing clinical conditions, available staff, diagnostic delays, and treatment capacity.

Proposed Innovation

- A clinician-assigned urgency system that categorizes patients based on their initial assessment and priority level.
- A real-time observation monitoring module that tracks changes in recorded vital signs and patient conditions.
- A resource status tracking system that monitors the availability of doctors, beds, equipment, and treatment facilities.
- A coordination engine that combines patient urgency, observation changes, and resource availability to identify cases requiring reassessment.
- A queue analysis mechanism that detects waiting-time delays, resource shortages, and bottlenecks in patient flow.
- A real-time dashboard that displays patient priorities, reassessment alerts, and resource status to support efficient clinical coordination.

Potential patent focus: A technical event-reconciliation mechanism that updates queue status when clinical and resource information changes asynchronously, without automatically replacing clinician triage decisions.

MEDAITHON
CODE 4 CURE
2026

MED-PS-011

REFERENCE

	Patent	Relevance to your concept	Link
1	**US8352285B2 – Dynamically Adjusting Triage Classification Levels**	Very close prior art: adjusts a new patient's triage level up or down using resource availability and the triage levels of other ED patients. Your concept differs by flagging for reassessment rather than replacing clinician triage.	Google Patents – US8352285B2
2	**US8484048B2 – Automated System and Method for Prioritization of Waiting Patients**	Algorithmic prioritisation of waiting patients using subjective and objective criteria with real-time status information.	Google Patents – US8484048B2
3	**US12148529B2 – E-Triage: An Electronic Emergency Triage System**	Risk-based electronic triage that includes predicted resource needs in the triage level.	Google Patents – US12148529B2
4	**US10839959 – System and Method and Graphical Interface for Performing Predictive Analysis and Prescriptive Remediation of Patient Flow and Care Delivery Bottlenecks Within Emergency Departments and Hospital Systems**	Monitors queue depth and time in queue, including resource queues (e.g., a busy CT scanner), with escalating alerts.	Google Patents – US10839959

5	**US8700425 – Dynamically Predicting**	Computes the current ED backlog from patient counts,	Google Patents – US8700425

	Patient Influx into an Emergency Department	triage levels and wait times to predict imminent influx.	

MED-PS-012

12.Critical Care / Intensive Care Unit

Problem Statement: Development of a Multidevice Alarm-Correlation System for Reducing Duplicate and Non-Actionable ICU Alerts

ICU teams receive alarms from multiple bedside devices. Related alarms may appear separately, making it difficult to distinguish one underlying event from several correlated notifications.

Proposed Innovation

- A bedside integration unit that collects and synchronizes alarms and physiological signals from multiple monitoring devices.
- A signal-quality analysis module that identifies unreliable readings and sensor disturbances.
- An event correlation engine that groups related alarms based on timestamps and signal patterns.
- A relationship analysis mechanism that identifies sequences of events and their possible connections.
- An explainable alert system that presents original signals, timestamps, and contributing events.
- A clinical dashboard that helps healthcare staff interpret related alarms and prioritize patient assessment.

Potential patent focus: A device-independent temporal correlation mechanism that groups alarms while preserving source-specific thresholds, alarm priority, and independent safety notifications.

REFERENCE

	Patent	Relevance to your concept	Link
1	**US10311696B2 – Patient Monitoring Method and System Providing Incident Grouping of Alarm Events**	Very close prior art: groups multiple alarm events into an incident group using a group alarm timer and an incident-group designator.	Google Patents – US10311696B2
2	**US9852265B1 – System and Method of Medical Device Alarm Management**	Receives alarm data from several medical devices and manages the alarms centrally.	Google Patents – US9852265B1
3	**US9700218B2 – Systems and Methods for Reducing Nuisance Alarms in Medical Devices**	Analyses alarm relevance data to modify alarm protocols and cut nuisance alarms.	Google Patents – US9700218B2
4	**US12131825B2 – Criteria Based Alarms Coordination Between a Network of Medical Devices**	Coordinates notifications between devices according to alarm type and priority.	Google Patents – US12131825B2
5	**US7936260B2 – Identifying Redundant Alarms by Determining Coefficients of Correlation Between Alarm Categories**	General (non-medical) statistical correlation of alarm categories to find redundant alarms with a common root cause.	Google Patents – US7936260B2

13. Cost-effective pelvic floor muscle stimulation for older women

Proposed title: Adaptive Pelvic Floor Rehabilitation and Biofeedback System for Older Women

Clinical need

Discipline: Biomedical engineering + embedded systems + rehabilitation

Problem statement: Older women experiencing urinary incontinence may face barriers to accessing pelvic floor rehabilitation, including cost, discomfort, difficulty performing exercises correctly and limited access to trained professionals. Develop an affordable rehabilitation system that supports individualized pelvic floor muscle training and provides measurable feedback on exercise performance.

Proposed innovation

- A wearable or external biofeedback interface that guides pelvic floor muscle exercises.
- A sensor system to measure appropriate exercise-related signals, such as validated muscle activity where feasible.
- An adaptive training interface that adjusts exercise prompts based on measured performance.
- A session log that helps a qualified clinician monitor progress.

Potential invention hypothesis: A specific closed-loop biofeedback method that adapts exercise guidance to validated muscle-activity measurements while detecting poor signal quality and compensatory muscle activation.

Prototype: Microcontroller, surface EMG acquisition module, reusable external sensors, mobile application and a training dashboard.

Validation: Begin with simulated signals and bench testing. Any clinical evaluation needs appropriate ethics approval and qualified clinical supervision.

Important: electrical stimulation is not suitable for every patient. Do not use a DIY electrical stimulator on participants. Patient-connected stimulation hardware requires electrical safety engineering, clinical oversight and appropriate medical-device regulatory assessment. A biofeedback-only prototype is the safer starting point.

REFERENCE

	Patent	Relevance to your concept	Link
1	**US11426115B2 – System and Method for Pelvic Floor Feedback and Neuromodulation**	Very close prior art: EMG biofeedback for the pelvic floor with sensor electrodes and neuromodulation; addresses electrode placement and equipment limits.	Google Patents – US11426115B2
2	**US11172866B2 – Pelvic Muscle Rehabilitation for Treating Urinary Incontinence**	Garment-integrated EMG sensor measuring pelvic floor contraction plus a moisture sensor, with stored outputs.	Google Patents – US11172866B2
3	**US10602979B2 – Method with Biofeedback for Training the Muscles of the Pelvic Floor**	Biofeedback training method based on EMG signals.	Google Patents – US10602979B2
4	**US6264582B1 – Method for Training of Muscles**	Intravaginal EMG probe, initial assessment of muscle state, auditory feedback and a positive-feedback training programme.	Google Patents – US6264582B1
5	**US6498953B2 – Biofeedback Device for an Incontinent Person**	Senses sphincter muscle electrical activity and signals on a predetermined activity to help the user contract the muscle.	Google Patents – US6498953B2

	Research	What it supports	Link
1	PelviSense, an Innovative, Non-invasive, Biofeedback Device for the Treatment of Stress Urinary Incontinence in Women (NCT06126757)	Registered trial of a reusable wearable EMG sensor with a mobile app showing real-time pelvic floor EMG, comparable to your prototype concept.	ClinicalTrials.gov – NCT06126757
2	Effectiveness and Cost-effectiveness of PelviSense, a Novel Biofeedback Device for Stress Urinary Incontinence in Women (NCT05627726)	Registered trial that includes cost-effectiveness and 7-day progress display in the app.	ClinicalTrials.gov – NCT05627726

14. Smart hand-hygiene compliance and technique verification

Proposed title: Privacy-Preserving Hand-Hygiene Opportunity and Technique Monitoring System

Discipline: IoT + sensor fusion + healthcare quality

Problem statement: Hand-hygiene adherence can vary across clinical areas and shifts. Conventional dispenser counters show product use but cannot reliably establish whether a hand-hygiene opportunity occurred or whether the required action was completed correctly. Develop a monitoring system that detects relevant workflow events, provides timely prompts and produces useful compliance data without continuous video surveillance or excessive staff workload.

Proposed innovation

- A dispenser sensor that records activation.
- A proximity or location sensor to identify relevant staff interactions.
- Optional door, bed-space or workflow-event sensors.
- A rules engine that estimates hand-hygiene opportunities and identifies possible missed events.
- A privacy-preserving dashboard that reports trends rather than unnecessary individual surveillance.

Potential invention hypothesis: An event-correlation method that distinguishes a relevant hand-hygiene opportunity from incidental staff proximity and dispenser activation, using confidence scores and privacy-preserving event records.

Prototype: ESP32, infrared or proximity sensors, dispenser activation sensor, optional RFID test badges and a web dashboard.

Validation: Compare system-generated events against manually observed, approved test scenarios. Measure false alerts, missed events, latency and staff interaction time.

Critical limitation: Dispenser activation is not proof of correct hand hygiene. A prototype must not claim to establish handwashing quality or infection prevention effectiveness unless those outcomes have been independently validated.

REFERENCE

	Patent	Relevance to your concept	Link
1	**US10475329B1 – Hand Washing Compliance and Monitoring System**	Very close prior art: dispenser activation + RFID worker identification + proximity sensor + compliance monitoring.	Google Patents – US10475329B1
2	**US20240161596A1 – System and Methods for Wireless Hand Hygiene Monitoring**	Extremely relevant: combines dispenser-event sensing, ultrasound proximity, unique staff identifiers, room entry/exit detection and alerts.	Google Patents – US20240161596A1
3	**US20190139395A1 – Hygiene Monitoring System**	Uses location sensors, predefined rules, patient zones, dispenser proximity and soap-dispenser events to determine whether hand hygiene occurred.	Google Patents – US20190139395A1
4	**US10748410B2 – System and Method for Detecting Hand Hygiene Compliance**	Uses proximity, dispenser sensors, infrared sensing and a wearable wristband to monitor hand movement during a hygiene event.	Google Patents – US10748410B2
5	**US20130033376A1 – Hand Hygiene Compliance System**	Uses wearable zone sensors, patient/entrance zones, timers, dispenser detection and prompts when a hygiene event may be required.	Google Patents – US20130033376A1

	Research	What it supports	Link
1	**Automated hand hygiene count devices may better measure compliance than human observation**	Shows that dispenser counts can differ substantially from actual observed compliance. This strongly supports your limitation that dispenser activation ≠ compliance.	PubMed – Automated hand hygiene count devices
2	**An electronic hand hygiene surveillance device: a pilot study exploring surrogate markers for hand hygiene compliance**	Demonstrates electronic monitoring using room-entry triggers and dispenser triggers, while treating these as surrogate measurements.	PubMed – Electronic hand hygiene surveillance device
3	**Evaluating the accuracy of an automatic counting system to detect dispensing of hand hygiene product**	Very relevant to your validation methodology: evaluates automatic dispenser counting and explicitly distinguishes product dispensing from actual Five Moments compliance.	PubMed – Automatic counting system accuracy
4	**Compliance and Performance of Hand Hygiene in Dutch General Practice Offices Using Electronic Dispensers**	Demonstrates electronic dispenser-event collection and comparison against estimated hand-hygiene opportunities derived from clinical workflow data.	PubMed – Electronic dispensers and hand hygiene compliance
5	**Observing is influencing: How hand disinfection compliance observations affect hand disinfection rates**	Recent research demonstrating the Hawthorne effect and differences between electronically detected events and direct observation. Useful for your privacy/non-intrusive monitoring rationale.	PubMed – Electronic monitoring and Hawthorne effect